

Symmetrical Components

→ Fortescue's Theorem

- 3 unbalanced phasors of a 3-phase system can be resolved into 3 balanced systems of phasors.
The balanced sets of components are:

① Positive-sequence components

- + 3 balanced phasors
 - ⇒ equal in magnitude
 - ⇒ displaced from each other by 120°
 - ⇒ same phase sequence as the original phasors (for example a-b-c)

② Negative-sequence components

- 3 balanced phasors
 - ⇒ equal in magnitude
 - ⇒ displaced from each other by 120°
 - ⇒ opposite phase sequence to the original phasors (for example a-c-b)

③ Zero-sequence components

- 0 3 equal phasors
 - ⇒ equal in magnitude
 - ⇒ zero phase displacement from each other

Symmetrical Components

Original voltages:

$$V_a \quad V_b \quad V_c$$

Positive-sequence components:

$$V_{a1} \quad V_{b1} \quad V_{c1}$$

or

$$V_a^+ \quad V_b^+ \quad V_c^+$$

Negative-sequence components:

$$V_{a2} \quad V_{b2} \quad V_{c2}$$

or

$$V_a^- \quad V_b^- \quad V_c^-$$

Zero-sequence components:

$$V_{a0} \quad V_{b0} \quad V_{c0}$$

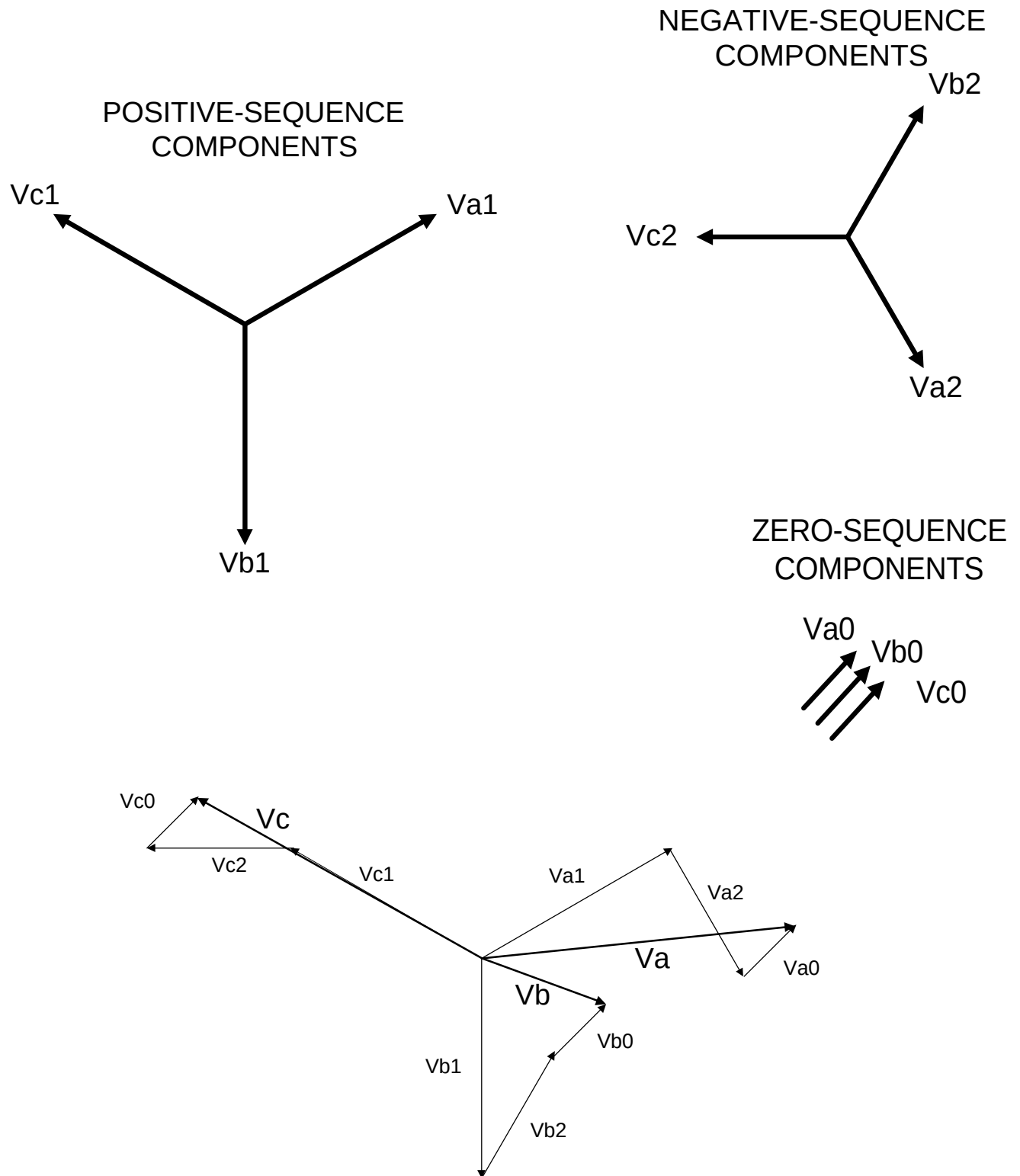
ORIGINAL PHASORS ARE THE SUM OF THEIR COMPONENTS:

$$V_a = V_{a0} + V_{a1} + V_{a2}$$

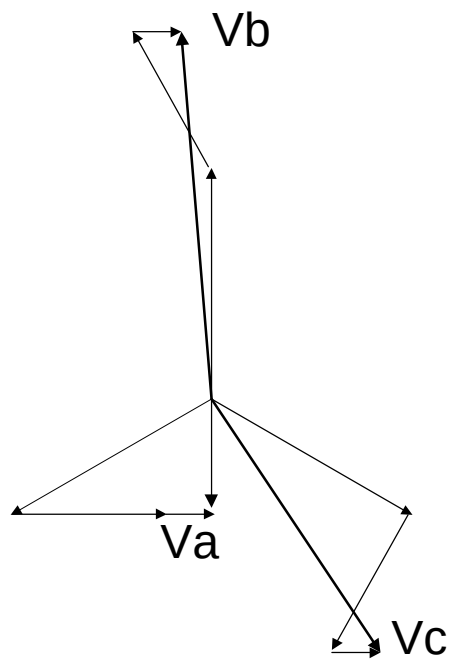
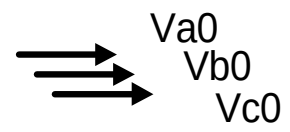
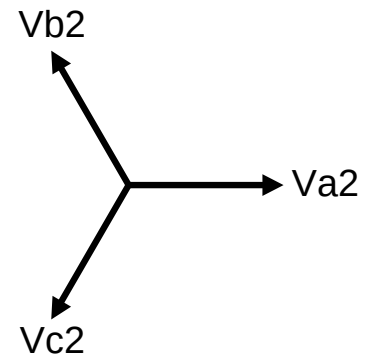
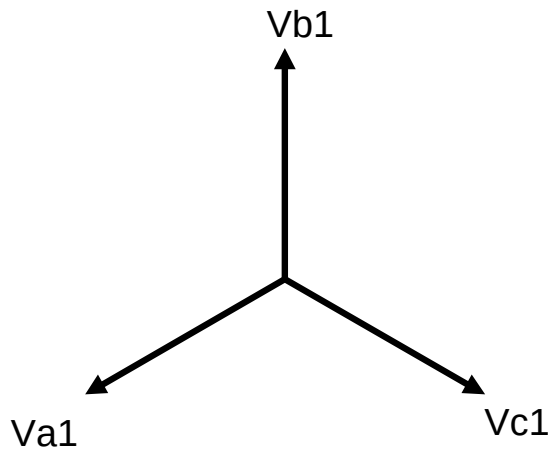
$$V_b = V_{b0} + V_{b1} + V_{b2}$$

$$V_c = V_{c0} + V_{c1} + V_{c2}$$

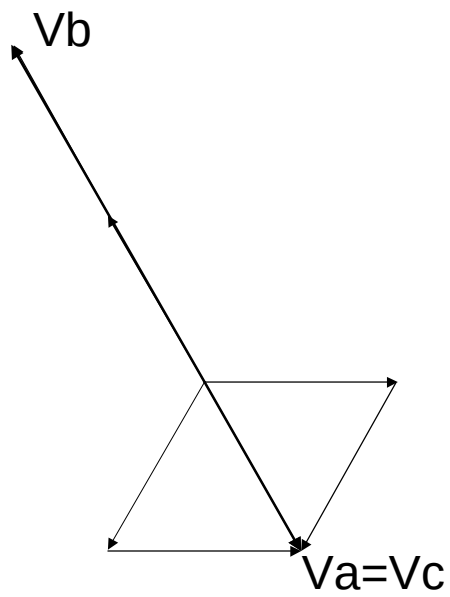
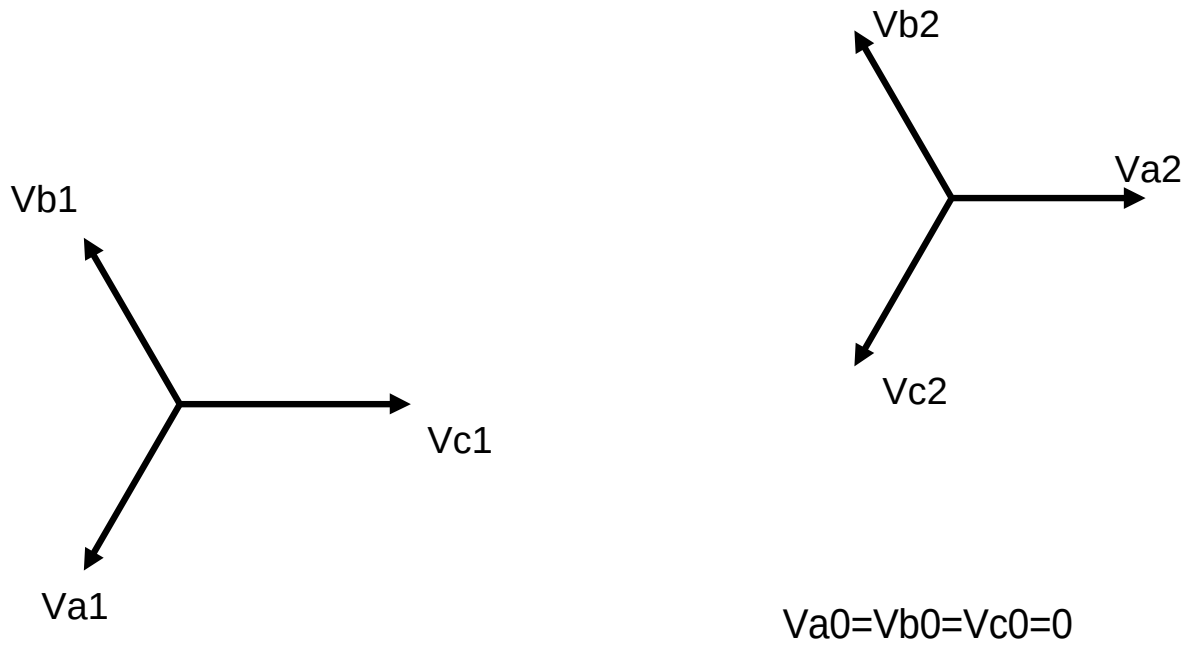
Example 1



Example 2



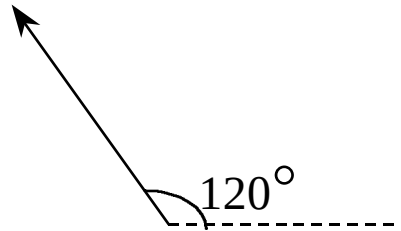
Example 3



Operators

→ Lets define a phasor:

$$\mathbf{a} = 1 \angle 120^\circ = -0.5 + j \, 0.8666$$

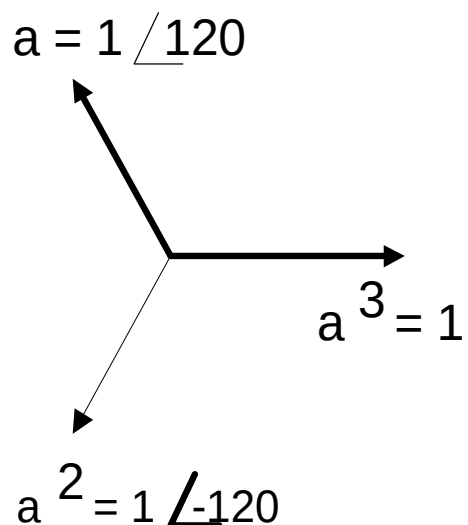


The following relations are true:

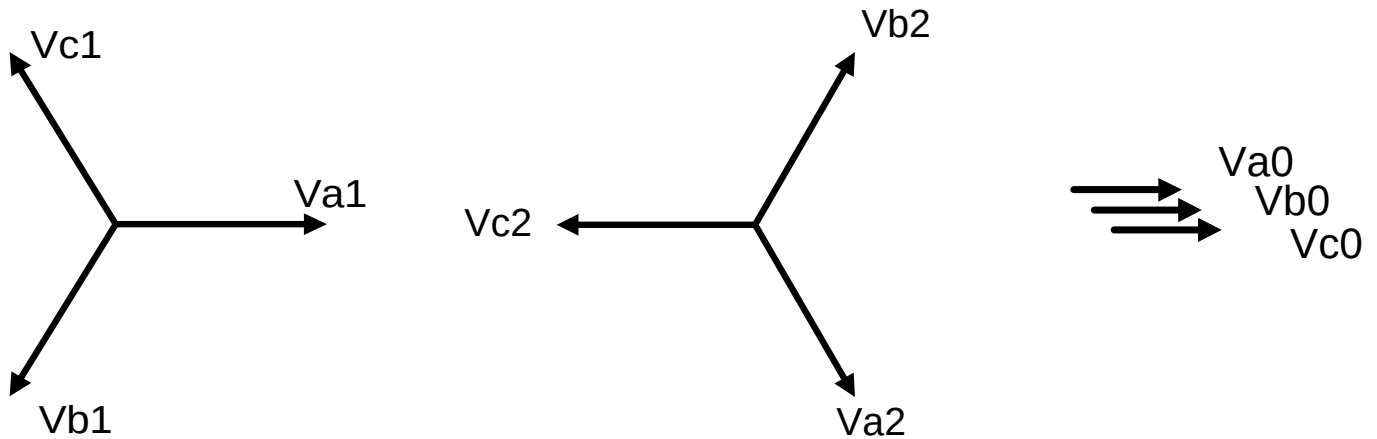
$$\mathbf{a}^2 = (1 \angle 120^\circ)(1 \angle 120^\circ) = 1 \angle 240^\circ = 1 \angle -120^\circ$$

$$\mathbf{a}^3 = 1 \angle 360^\circ = 1 \angle 0^\circ$$

$$1 + \mathbf{a} + \mathbf{a}^2 = 0$$



Symmetrical Components Relations



The positive-sequence components can be written as:

$$V_{b1} = (1 \angle -120^\circ) V_{a1} = \mathbf{a}^2 V_{a1}$$

$$V_{c1} = (1 \angle 120^\circ) V_{a1} = \mathbf{a} V_{a1}$$

For the negative-sequence, we have:

$$V_{b2} = (1 \angle 120^\circ) V_{a2} = \mathbf{a} V_{a2}$$

$$V_{c2} = (1 \angle -120^\circ) V_{a2} = \mathbf{a}^2 V_{a2}$$

And for the zero-sequence:

$$V_{a0} = V_{b0} = V_{c0}$$

The totals are:

$$V_a = V_{a0} + V_{a1} + V_{a2}$$

$$V_b = V_{b0} + V_{b1} + V_{b2}$$

$$V_c = V_{c0} + V_{c1} + V_{c2}$$

$$= V_{a0} + V_{a1} + V_{a2}$$

$$= V_{a0} + \mathbf{a}^2 V_{a1} + \mathbf{a} V_{a2}$$

$$= V_{a0} + \mathbf{a} V_{a1} + \mathbf{a}^2 V_{a2}$$

Symmetrical Components Relations

In matrix form :

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix}$$

This defines the transformation matrix :

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix}$$

Its inverse is :

$$A^{-1} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}$$

Therefore :

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix}$$

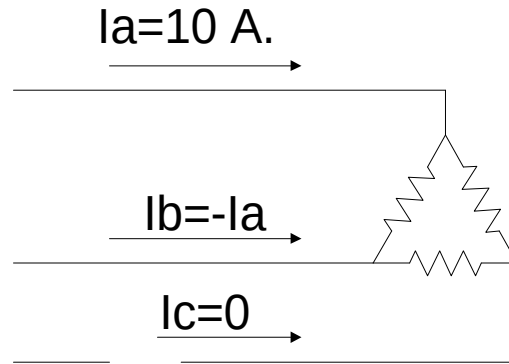
or :

$$V_{a0} = \frac{1}{3} (V_a + V_b + V_c)$$

$$V_{a1} = \frac{1}{3} (V_a + a V_b + a^2 V_c)$$

$$V_{a2} = \frac{1}{3} (V_a + a^2 V_b + a V_c)$$

Example



$$I_a = 10 \angle 0^\circ \text{ A.} \quad I_b = 10 \angle 180^\circ \text{ A.} \quad I_c = 0 \text{ A.}$$

$$I_{a0} = \frac{1}{3} [10 \angle 0^\circ + 10 \angle 180^\circ + 0] = 0$$

$$I_{a1} = \frac{1}{3} [10 \angle 0^\circ + (10 \angle 180^\circ)(1 \angle 120^\circ) + 0] = 5.78 \angle -30^\circ$$

$$I_{a2} = \frac{1}{3} [10 \angle 0^\circ + (10 \angle 180^\circ)(1 \angle -120^\circ) + 0] = 5.78 \angle 30^\circ$$

$$I_{b0} = I_{a0} = 0$$

$$I_{b1} = 5.78 \angle -30^\circ - 120^\circ = 5.78 \angle -150^\circ$$

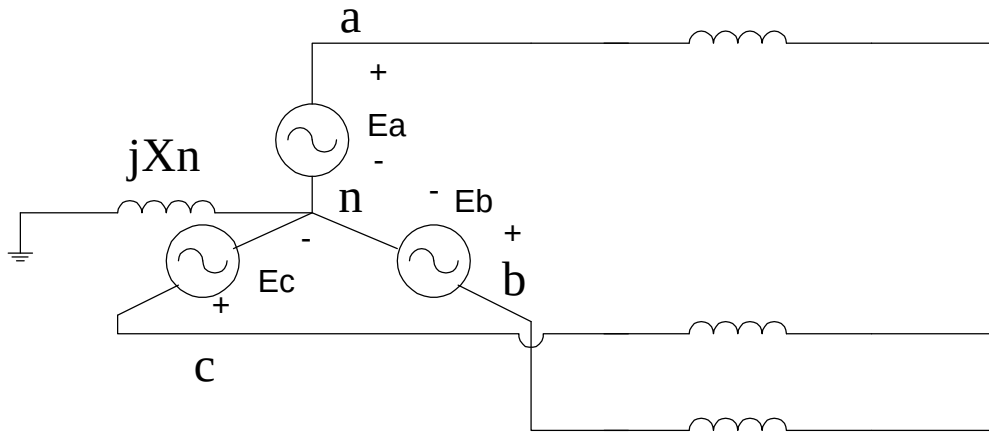
$$I_{b2} = 5.78 \angle +30^\circ + 120^\circ = 5.78 \angle +150^\circ$$

$$I_{c0} = I_{a0} = 0$$

$$I_{c1} = 5.78 \angle -30^\circ + 120^\circ = 5.78 \angle +90^\circ$$

$$I_{c2} = 5.78 \angle +30^\circ - 120^\circ = 5.78 \angle -90^\circ$$

Unloaded Generator



Generator is grounded through a grounding reactor

Internal voltages are balanced.

Find the symmetrical components for the internal voltages.

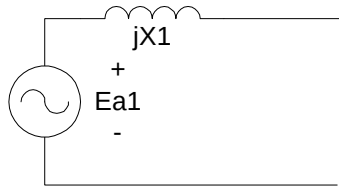
$$\begin{bmatrix} E_{a0} \\ E_{a1} \\ E_{a2} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} E_a \\ E_b \\ E_c \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} E_a \\ a^2 E_a \\ a E_a \end{bmatrix}$$

$$\begin{bmatrix} E_{a0} \\ E_{a1} \\ E_{a2} \end{bmatrix} = \begin{bmatrix} 0 \\ E_a \\ 0 \end{bmatrix}$$

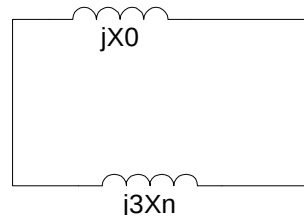
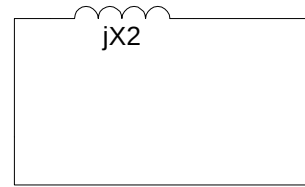
Only positive-sequence voltage exists!!!

Generator Equivalent

Positive-sequence



Negative-sequence



Zero-sequence
 X_n = Impedance from
 neutral to ground

The current in the neutral of the generator is:

$$I_n = I_a + I_b + I_c = (I_{a1} + I_{b1} + I_{c1}) + (I_{a2} + I_{b2} + I_{c2}) + I_{a0} + I_{b0} + I_{c0}$$

The positive and negative sequence components add to zero:

$$I_{a1} + I_{b1} + I_{c1} = 0$$

$$I_{a2} + I_{b2} + I_{c2} = 0$$

This means that the neutral does not carry positive or negative sequence components.

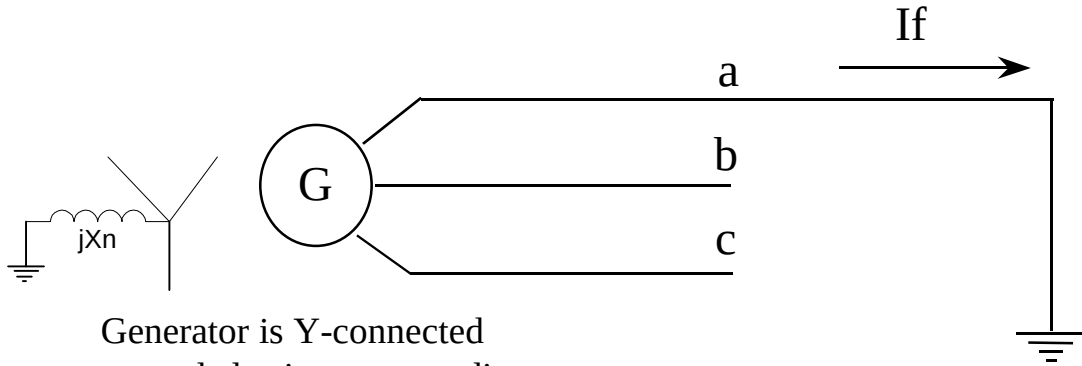
However, the zero-sequence components are in phase, and their sum is:

$$I_n = I_{a0} + I_{b0} + I_{c0} = 3 I_{a0}$$

Therefore the zero sequence equivalent has a grounding impedance of value: $Z_g = 3 j X_n$

Single-line to ground fault

Unloaded generator (with balanced internal voltages)



Generator is Y-connected
grounded using a grounding reactor

$$I_a = I_f \quad V_a = 0$$

$$I_b = 0 \quad V_b = ?$$

$$I_c = 0 \quad V_c = ?$$

$$\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} I_f \\ 0 \\ 0 \end{bmatrix}$$

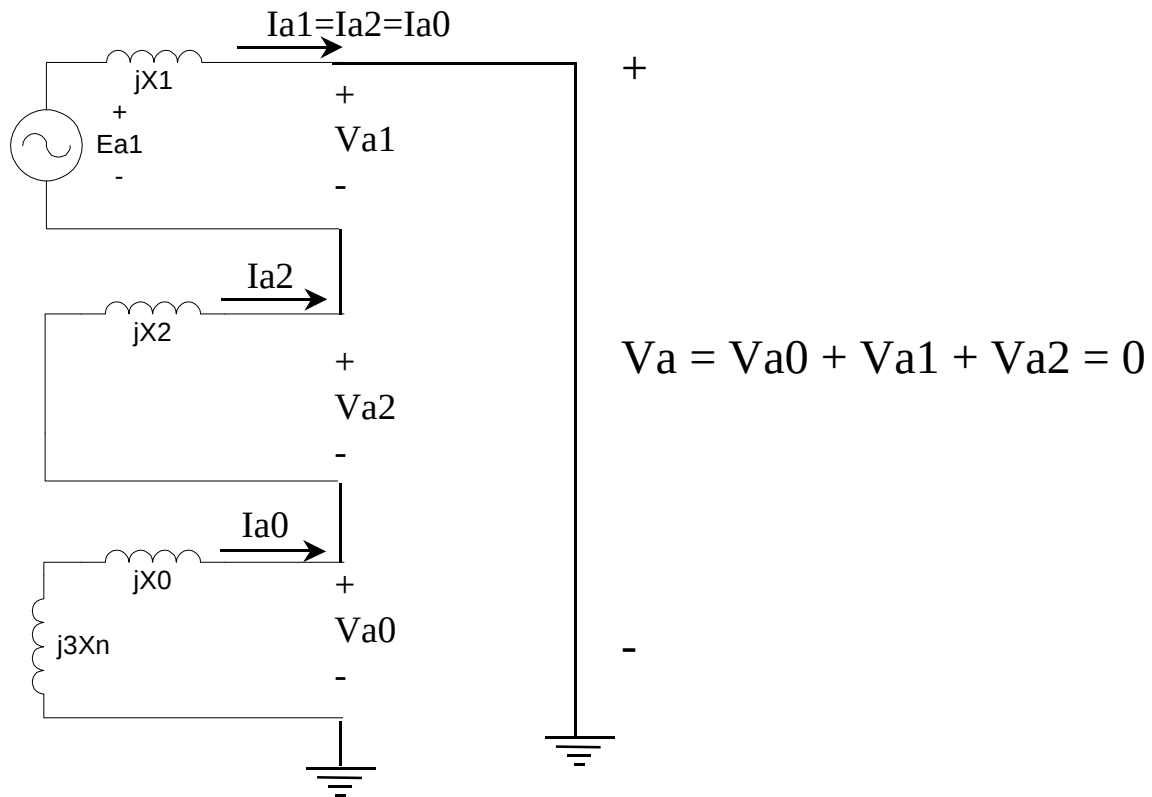
$$\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \begin{bmatrix} I_f / 3 \\ I_f / 3 \\ I_f / 3 \end{bmatrix}$$

$$I_{a0} = I_{b0} = I_{c0} = \frac{I_f}{3}$$

Single-line to ground fault

We have $I_{a1}=I_{a2}=I_{a0}$ and $V_a=0$.

This situation can be represented in the following way:



From the circuit:

$$I_{a0} = I_{a1} = I_{a2} = \frac{E_{a1}}{jX_1 + jX_2 + jX_0 + j3X_n}$$

$$V_{a1} = E_{a1} - (jX_1)(I_{a1})$$

$$V_{a2} = 0 - (jX_2)(I_{a2})$$

$$V_{a0} = 0 - j(X_0 + 3X_n)(I_{a0})$$

With this term
we include the
value of the
grounding
reactor

Example: Line-to-ground fault in an unloaded generator

Assume:

$$X_1 = X_2 = 0.12 \text{ p.u.} \quad X_0 = 0.06 \text{ p.u.}, \quad X_n = 0 \quad E_{a1} = 1 \text{ p.u.}$$

$$I_{a0} = I_{a1} = I_{a2} = \frac{1 \angle 0^\circ}{j(0.12 + 0.12 + 0.06)} = -j3.33 \text{ p.u.}$$

$$I_a = I_f = 3I_{a1} = -j10 \text{ p.u.}$$

$$I_b = 0$$

$$I_c = 0$$

$$V_{a0} = -jX_0(I_{a0}) = -j0.06(-j3.33) = -0.2$$

$$V_{a1} = E_{a1} - jX_1(I_{a1}) = 1 - j0.12(-j3.33) = 1 - j0.4 = 0.6$$

$$V_{a2} = -jX_2(I_{a2}) = -j0.12(-j3.33) = -0.4$$

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix}$$

$$V_a = V_{a0} + V_{a1} + V_{a2} = -0.2 + 0.6 - 0.4 = 0$$

$$V_b = V_{a0} + a^2 V_{a1} + a V_{a2}$$

$$V_b = -0.2 + 0.6a^2 - 0.4a = -0.2 + 0.6 \angle -120^\circ - 0.4 \angle 120^\circ$$

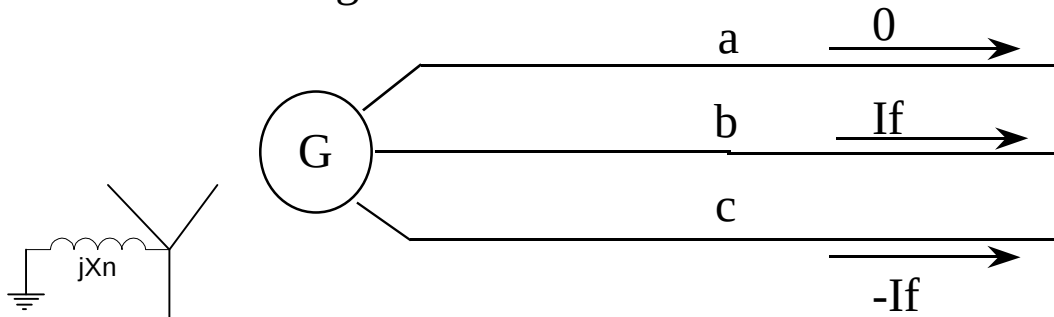
$$V_b = 0.9165 \angle -109.1^\circ$$

$$V_c = V_{a0} + a V_{a1} + a^2 V_{a2} = -0.2 + 0.6 \angle 120^\circ - 0.4 \angle -120^\circ$$

$$V_c = 0.9165 \angle +109.1^\circ$$

Line-to-line fault

Unloaded generator



$$I_a = 0$$

$$V_a = ?$$

$$I_b = I_f$$

$$V_b = V_c$$

$$I_c = -I_f$$

$$\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} 0 \\ I_f \\ -I_f \end{bmatrix}$$

$$\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \frac{I_f}{3} \begin{bmatrix} 1 - 1 \\ a - a^2 \\ a^2 - a \end{bmatrix}$$

$$I_{a0} = 0$$

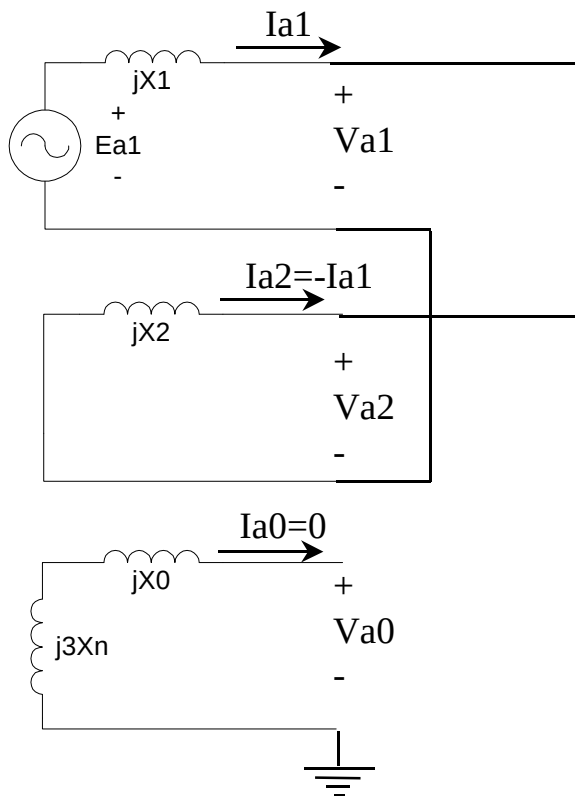
$$I_{a1} = j \left(\frac{I_f}{\sqrt{3}} \right)$$

$$I_{a2} = -j \left(\frac{I_f}{\sqrt{3}} \right)$$

Line-to-line fault

We have $I_{a1} = -I_{a2}$, $I_{a0} = 0$.

This situation can be represented in the following way:



From the circuit:

$$I_{a1} = -I_{a2} = \frac{E_{a1}}{jX_1 + jX_2}$$

$$V_{a1} = E_{a1} - (jX_1)(I_{a1})$$

$$V_{a2} = V_{a1}$$

$$V_{a0} = 0$$

Example: Line-to-line fault in an unloaded generator

Assume:

$$X_1 = X_2 = 0.12 \text{ p.u.} \quad X_0 = 0.06 \text{ p.u.}, \quad X_n = 0 \quad E_{a1} = 1 \text{ p.u.}$$

$$I_{a1} = -I_{a2} = \frac{1 \angle 0^\circ}{j(0.12 + 0.12)} = -j4.17 \text{ p.u.}$$

$$I_{a0} = 0$$

Phase currents during the fault are:

$$I_a = 0$$

$$I_b = I_f = -j\sqrt{3} I_{a1} = -7.22 \text{ p.u.}$$

$$I_c = -I_f = 7.22 \text{ p.u.}$$

$$V_{a0} = -jX_0(I_{a0}) = 0$$

$$V_{a1} = E_{a1} - jX_1(I_{a1}) = 1 - j0.12(-j4.17) = 1 - 0.5 = 0.5$$

$$V_{a2} = -jX_2(I_{a2}) = -j0.12(j4.17) = 0.5$$

Phase voltages during the fault are:

$$V_a = V_{a0} + V_{a1} + V_{a2}$$

$$V_a = 0 + 0.5 + 0.5 = 1.0 \text{ p.u.} \quad \text{Did not change!!!}$$

$$V_b = V_{a0} + a^2 V_{a1} + a V_{a2}$$

$$V_b = 0 + 0.5a^2 + 0.5a = 0.5(a^2 + a) = 0.5(-1)$$

$$V_b = -0.5 \text{ p.u.}$$

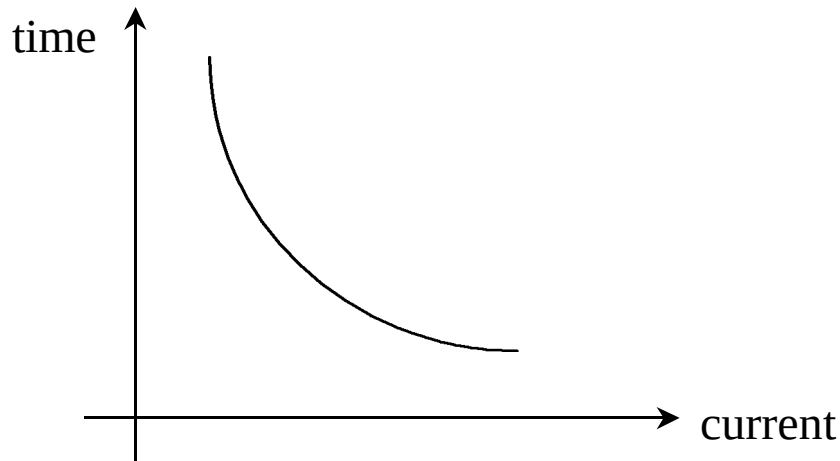
$$V_c = V_{a0} + a V_{a1} + a^2 V_{a2}$$

$$V_c = 0.5(a + a^2) = -0.5$$

$$V_c = -0.5 \text{ p.u.}$$

Protection in Industrial Electric Networks

- Objective: Isolation of the problem with the minimum service disruption
- Based on the protective device Time-Current characteristics



- Information needed for a protection study:
 - ☐ Protective device manufacturer and type
 - ☐ Protective device ratings
 - ☐ Trip settings and ratings
 - ☐ Short circuit current at each bus
 - ☐ Full load current of all loads
 - ☐ Voltage level at each bus
 - ☐ Power transformers data
 - ☐ Instrumentation transformers ratios

Coordination of the Protection